GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS
PHASE 5: Project Development Phase

Project theme: historical Olympic participation and medal performance.

Objective

Implement the data preparation, analysis, and visualization workflow.

Development steps

- Load the CSV with Pandas and inspect the schema.
- Check data types, missing values, category spelling, and year coverage.
- Prepare analysis columns and save the reduced CSV.
- Group records by Year, Season, NOC, Sport, and Medal as required.
- Generate tables for medals by NOC, medals by sport, and Olympic performance by year.
- Create charts and save the results.
- Write concise interpretations based on the computed results.

Core analytical measures

- Participation entries: number of athlete-event rows for a chosen group.
- Medal records: number of rows where Medal is Gold, Silver, or Bronze.
- Medals by NOC/team: medal records grouped by NOC/team and Olympic edition.
- Medals by sport: medal records grouped by sport.
- Trend measures: counts per Olympic year or edition.

Implementation safeguards

- Do not count blank Medal values as medals.
- Do not treat athlete-event rows as unique-person counts.
- Document whether medal counts are row counts or deduplicated awards.
- Keep raw data unchanged and save processed outputs separately.
- Use reproducible code and consistent grouping rules.

Deliverables

Analysis-ready dataset, generated summary CSVs, charts, and code/notebook with explanatory notes.

Expected phase deliverable

Reproducible data processing and analysis outputs.